Theory of the Standard Error Ellipse

In the case of two dimensions (x and y) the joint probability density function of a bivariate normal distribution is given by:

$$f(x,y) = \frac{1}{2\pi\sigma_x\sigma_y * \sqrt{(1-\rho_{xy}{}^2)}} * e^{\frac{-1}{2(1-\rho_{xy}{}^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 - 2\rho_{xy}\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right)+\left(\frac{y-\mu_y}{\sigma_y}\right)^2\right]}$$

Where:

- μ_x, μ_y: mean coordinates
- σ_x, σ_y: variances
- ρ_{xy}: correlation coefficient between x and y (Defined as $\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x\sigma_y}$ where σ_{xy} the covariance between x and y.)

Based on ***Figure 1***, the intersection of the joint probability density surface with a plane parallel to the xy-plane at a given height forms a contour, which corresponds to an error ellipse at a specific confidence level.

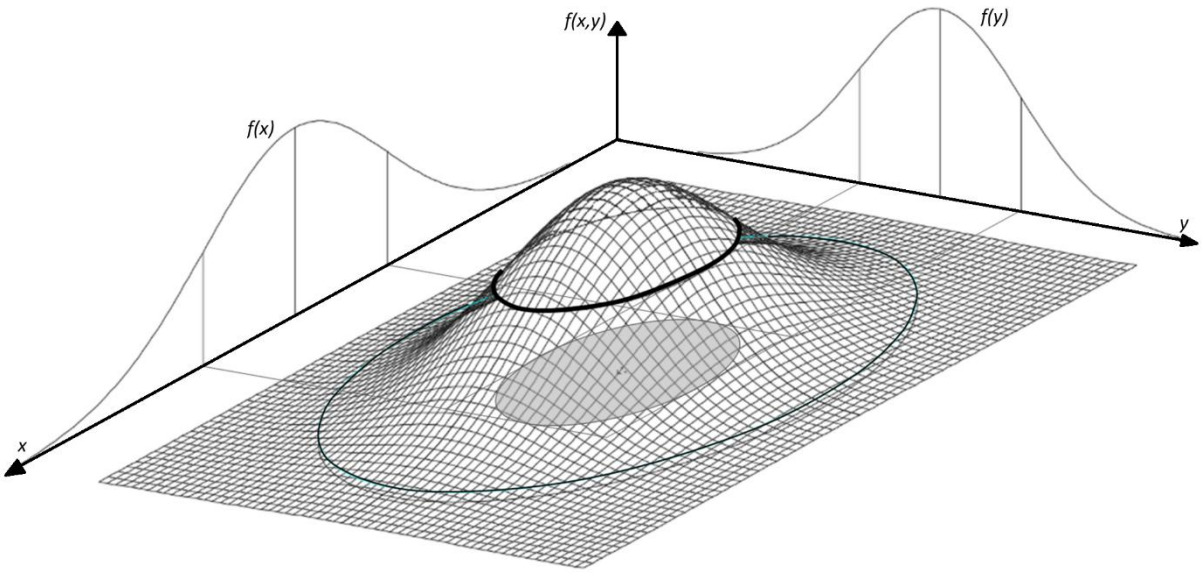

Figure 1: Visualization of the joint probability density function of a bivariate normal distribution (source: Amerisurv.com, modified)

By defining k as a constant corresponding to a specific confidence level, the equation of the error ellipse is given by:

$$\left(\frac{x-\mu_x}{\sigma_x}\right)^2 - 2\rho_{xy}\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right) + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 = (1-{\rho_{xy}}^2)c^2$$

Where:

$$c^2 = \ln\left[4\pi^2 k^2 {\sigma_x}^2 {\sigma_y}^2 (1-{\rho_{xy}}^2)\right]^{-1}$$

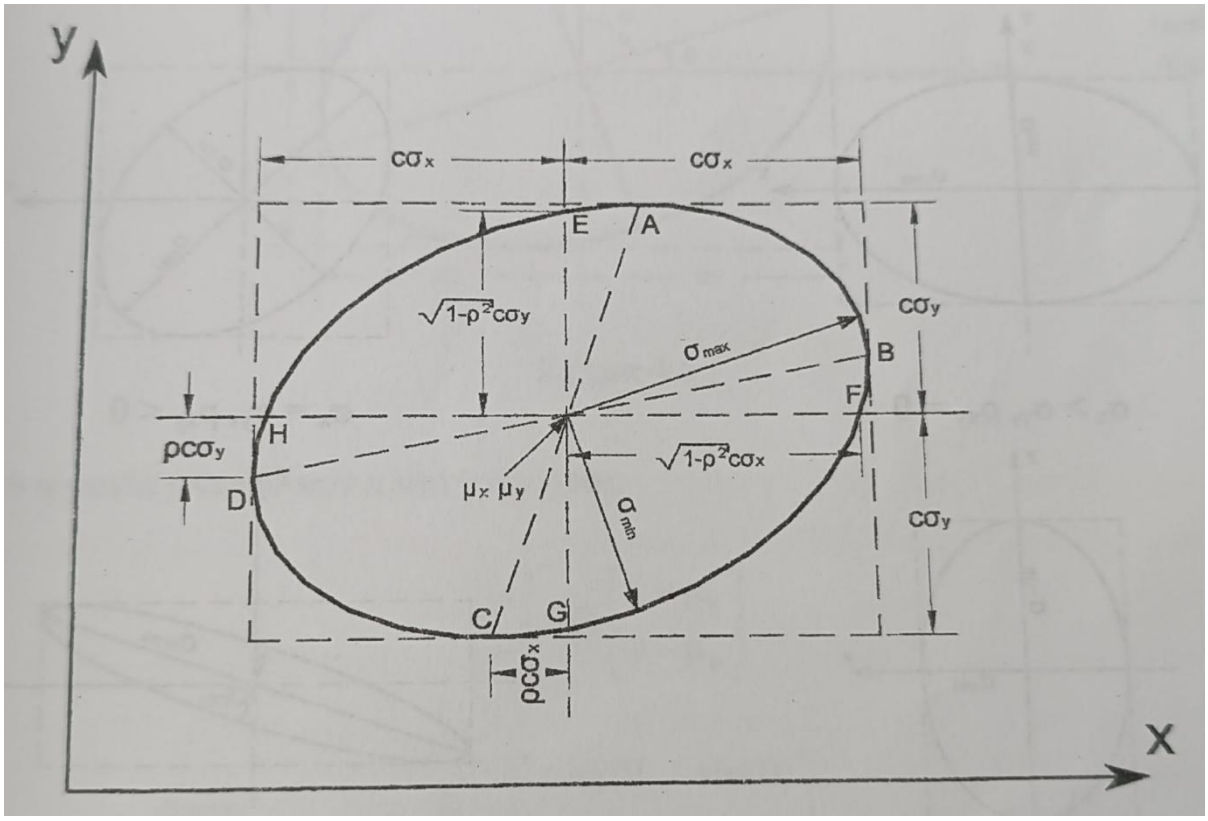

Figure 2: First visualization of error ellipse (source: A.M. Agatza-Balodimou, 2009)

According to ***Figure 2***, the error ellipse is inscribed within a rectangle centered at (μ_x, μ_y), with side lengths $2c\sigma_x$ and $2c\sigma_y$. **For $c = 1$**, this corresponds to the **standard error ellipse**, which represents the two-dimensional positional uncertainty of a surveyed point. In this case, **the probability that the true point position lies within the ellipse is approximately 39.4%.**

To calculate the error ellipse parameters:

- Major axis (σ_{max}):

$$\sigma_{max} = \sigma_u = \sqrt{\frac{1}{2}\left[\left(\sigma_x{}^2 + \sigma_y{}^2\right) + \sqrt{\left(\sigma_x{}^2 - \sigma_y{}^2\right)^2 + 4\sigma_{xy}{}^2}\right]}$$

- Minor axis (σ_{min}):

$$\sigma_{min} = \sigma_v = \sqrt{\frac{1}{2}\left[\left(\sigma_x{}^2 + \sigma_y{}^2\right) - \sqrt{\left(\sigma_x{}^2 - \sigma_y{}^2\right)^2 + 4\sigma_{xy}{}^2}\right]}$$

- Counterclockwise angle between the ellipse major axis $u$and the x-axis (θ):

$$\tan(2\theta) = \frac{2\sigma_{xy}}{\sigma_x{}^2 - \sigma_y{}^2}$$

Depending on the values of σ_x, σ_y, and σ_{xy}, the final angle θ may need to be adjusted as follows:

$\sigma_x > \sigma_y$	$\sigma_{xy} > 0$	$\theta_{final} = \theta$
$\sigma_x > \sigma_y$	$\sigma_{xy} < 0$	$\theta_{final} = \theta + 200^g$
$\sigma_x < \sigma_y$	$\sigma_{xy} > 0$	$\theta_{final} = \theta + 100^g$
$\sigma_x < \sigma_y$	$\sigma_{xy} < 0$	$\theta_{final} = \theta + 100^g$
$\sigma_x = \sigma_y$	$\sigma_{xy} > 0$	$\theta_{final} = 50^g$
$\sigma_x = \sigma_y$	$\sigma_{xy} < 0$	$\theta_{final} = 150^g$

Next the rotation matrix of the error ellipse, $\boldsymbol{R}$, is calculated as follows:

$$\boldsymbol{R} = \begin{bmatrix} \cos(\theta_{final}) & \sin(\theta_{final}) \\ -\sin(\theta_{final}) & \cos(\theta_{final}) \end{bmatrix}$$

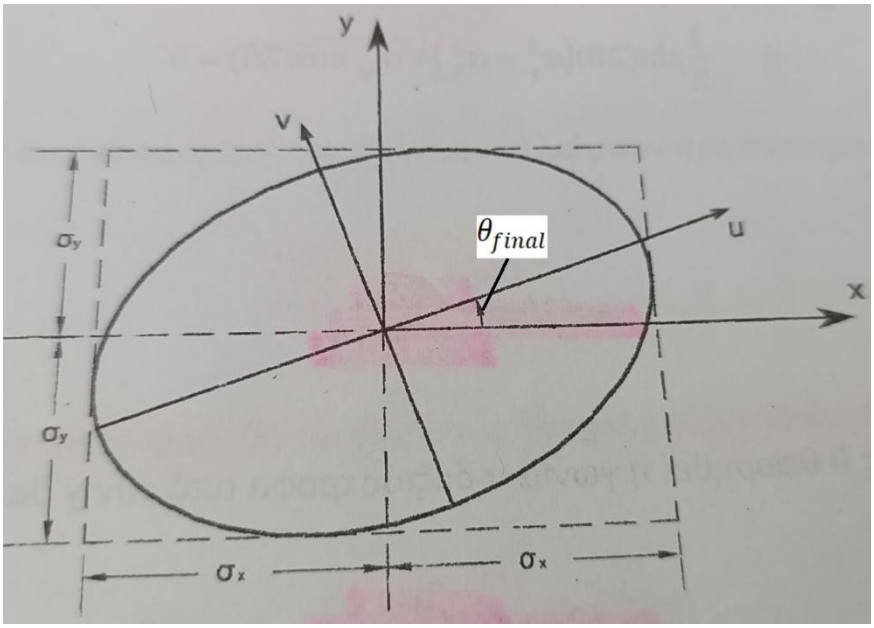

Figure 3: Second visualization of error ellipse (source: A.M. Agatza-Balodimou, 2009)